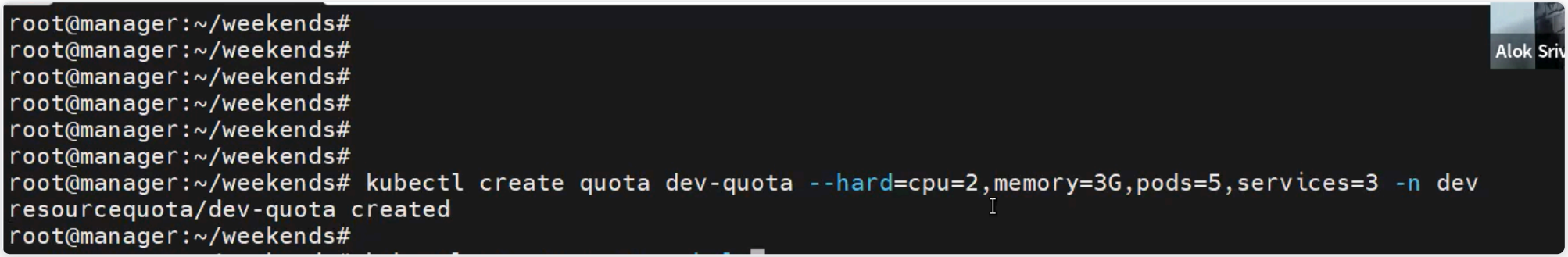
KUBERNETES CKA PREP › MY LEARNING NOTES › SESSION 05

Screenshots: Namespace Quota & LimitRange

Raw screenshots captured while watching this session, kept alongside the written notes (05-namespace-quota-and-limitrange.md) for visual reference during interview revision.

11 screenshots · captured 2026-09-11, 5:26 PM - 6:02 PM


root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# kubectl create quota dev-quota --hard=cpu=2,memory=3G,pods=5,services=3 -n dev
resourcequota/dev-quota created
root@manager:~/weekends#
Alok Sriv

```
root@manager:~/weekends#
root@manager:~/weekends# kubectl create quota dev-quota --hard=cpu=2,memory=3G,pods=5,services=3 -n dev
resourcequota/dev-quota created
root@manager:~/weekends#
root@manager:~/weekends# kubectl describe ns dev
Name:         dev
Labels:       area=dev
              kubernetes.io/metadata.name=dev
Annotations:  <none>
Status:       Active

Resource Quotas
  Name:      dev-quota
  Resource  Used  Hard
  --------  ---   ---
  cpu       0     2
  memory    0     3G
  pods      1     5
  services  0     3

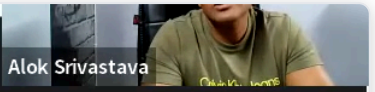
No LimitRange resource.
```

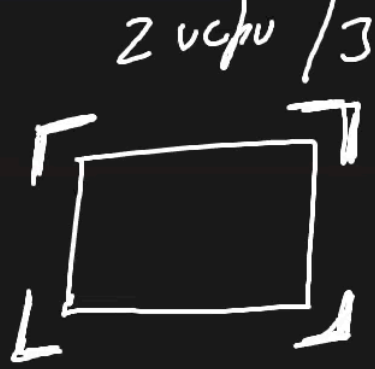
```
  Name:         dev-quota
  Resource      Used  Hard
  --------      ---   ---
  cpu           0     2
  memory        0     3G
  pods          0     5
  services      0     3

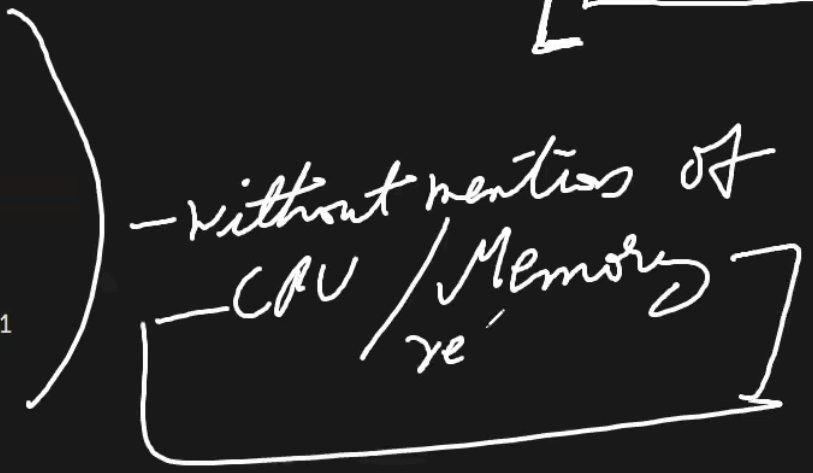
No LimitRange resource.
root@manager:~/weekends# cat mypod.yaml
---
apiVersion: v1
kind: Pod
metadata:
  name: webapp
  labels:
    app: web
    client: xyz
    env: prod
spec:
  containers:
  - name: nginx
    image: lovelearnlinux/webserver:v1
    ports:
    - containerPort: 80

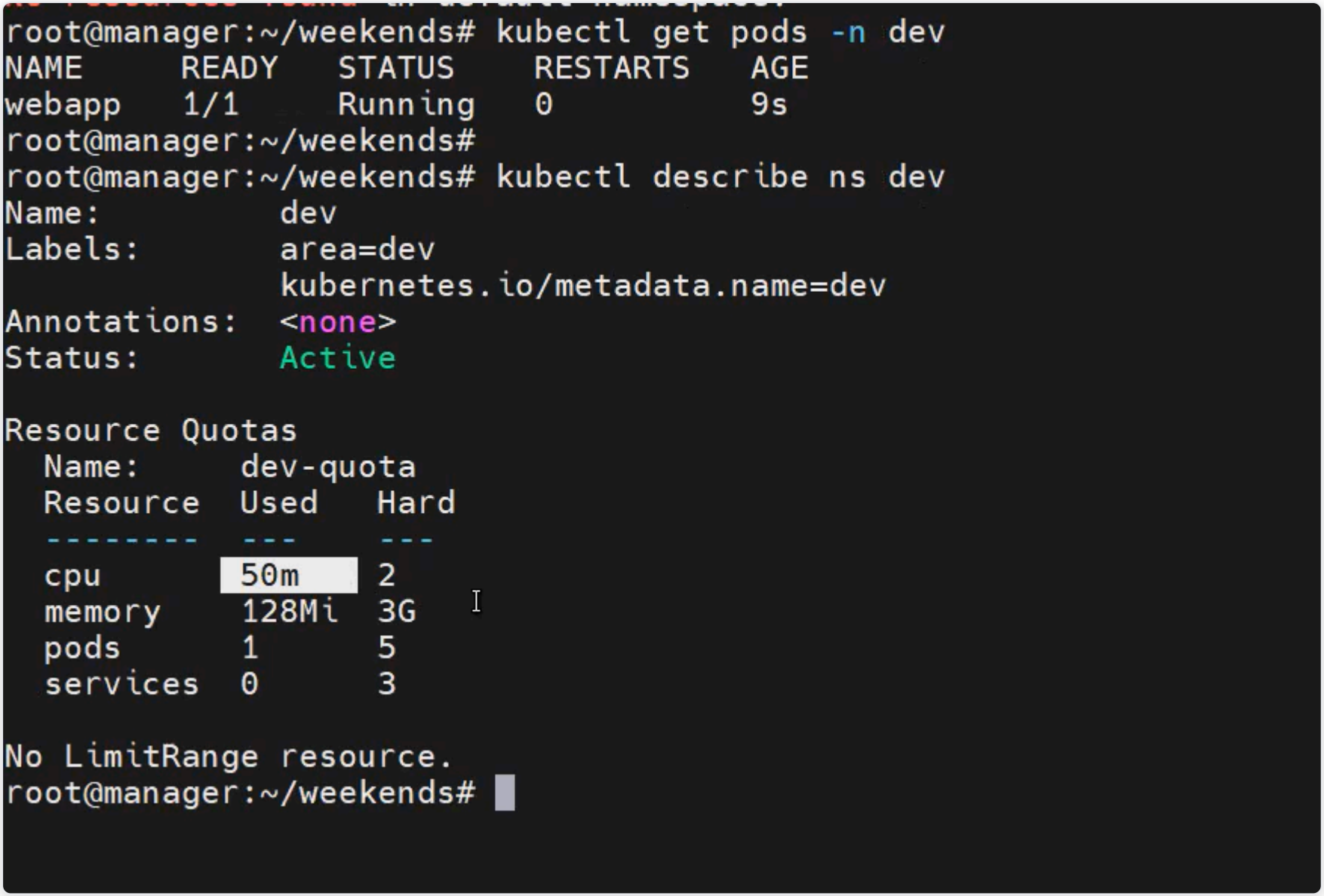


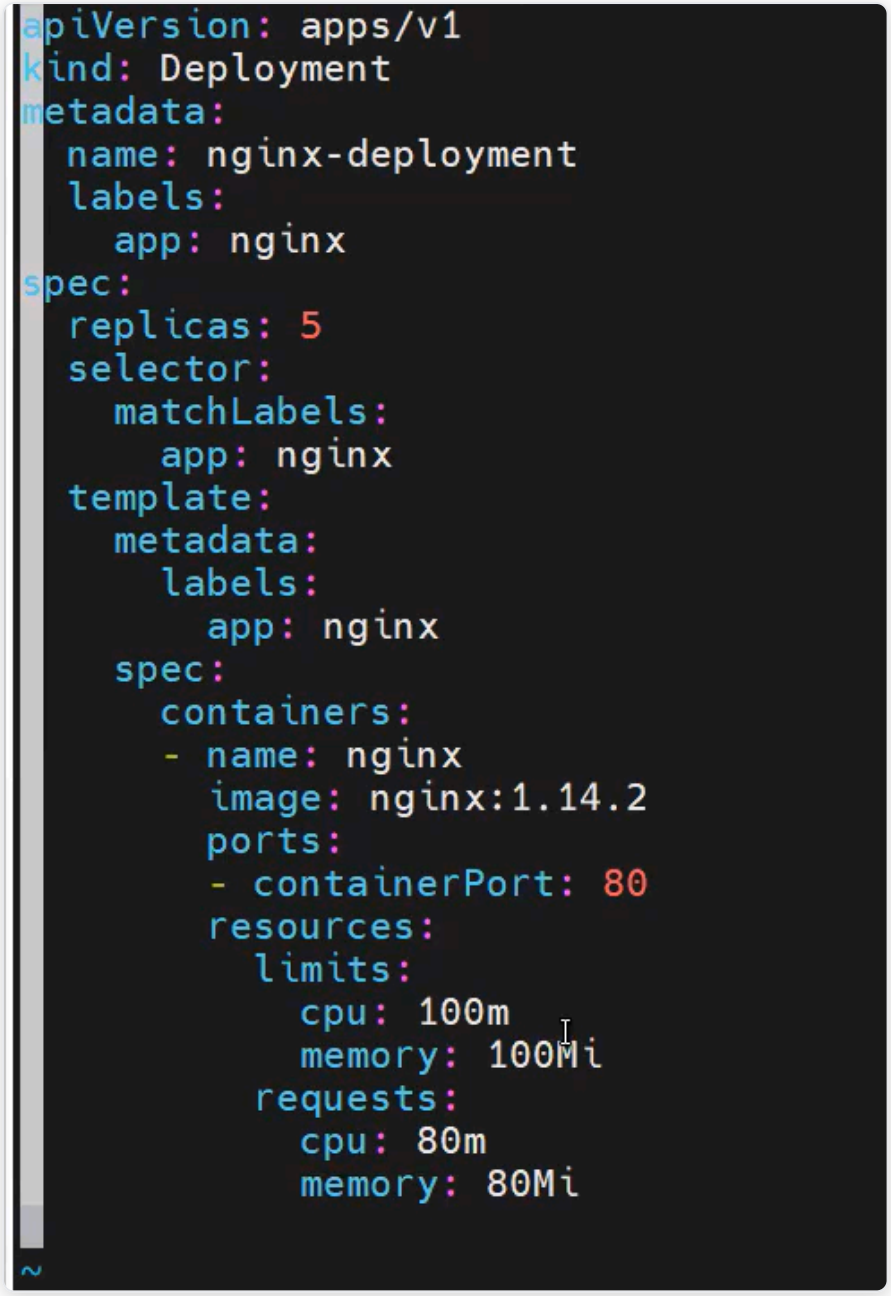
root@manager:~/weekends# vim mypod.yaml
root@manager:~/weekends#
root@manager:~/weekends# kubectl create -f mypod.yaml
Error from server (Forbidden): error when creating "mypod.yaml": pods "webapp" is forbidden: failed quota: dev-quota: must spe
cify cpu for: nginx; memory for: nginx
```








root@manager:~/weekends# kubectl get pods -n dev
NAME READY STATUS RESTARTS AGE
webapp 1/1 Running 0 9s
root@manager:~/weekends#
root@manager:~/weekends# kubectl describe ns dev
Name: dev
Labels: area=dev
kubernetes.io/metadata.name=dev
Annotations: <none>
Status: Active
Resource Quotas
Name: dev-quota
Resource Used Hard
cpu 50m 2
memory 128Mi 3G
pods 1 5
services 0 3
No LimitRange resource.
root@manager:~/weekends#



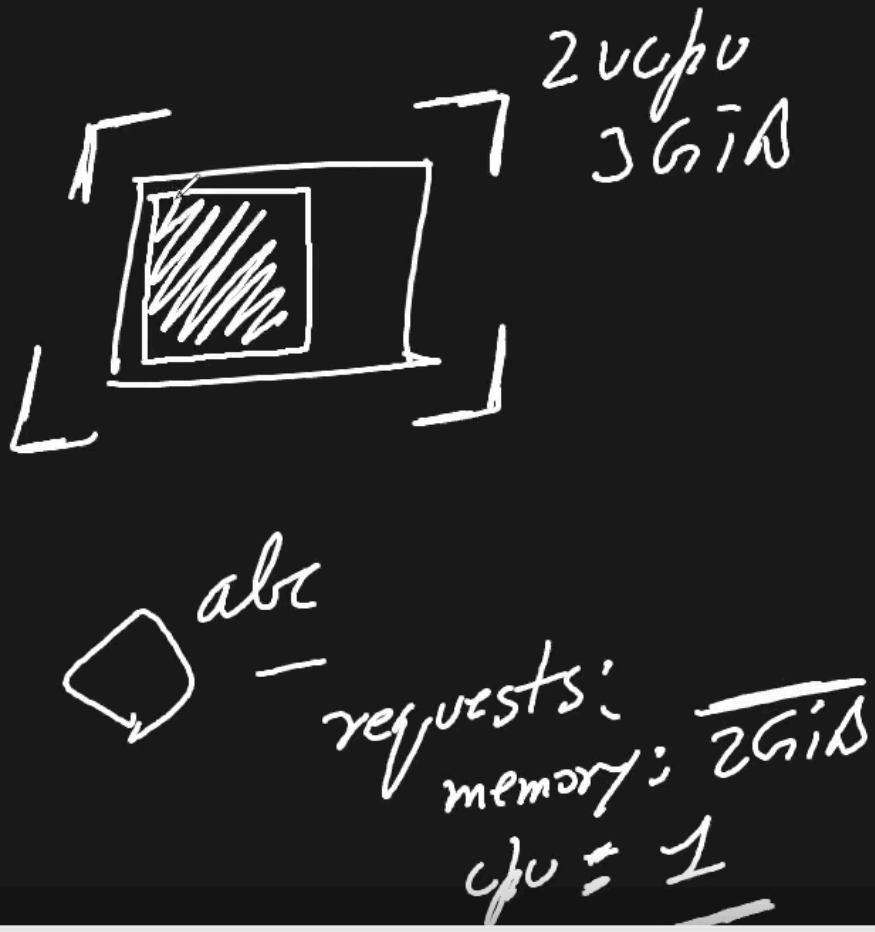
```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 5
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:1.14.2
        ports:
        - containerPort: 80
        resources:
          limits:
            cpu: 100m
            memory: 100Mi
          requests:
            cpu: 80m
            memory: 80Mi
~
```

```
  Type             Status  Reason
  ----             ------  ------
  ReplicaFailure   True    FailedCreate
Events:
  Type     Reason            Age                From                   Message
  ----     ------            ----               ----                   -------
  Normal   SuccessfulCreate  27s                replicaset-controller  Created pod: nginx-deployment-6bcdfcf6cc-nn7wz
  Normal   SuccessfulCreate  27s                replicaset-controller  Created pod: nginx-deployment-6bcdfcf6cc-wdt72
  Normal   SuccessfulCreate  27s                replicaset-controller  Created pod: nginx-deployment-6bcdfcf6cc-9q2dq
  Warning  FailedCreate      26s                replicaset-controller  Error creating: pods "nginx-deployment-6bcdfcf6cc-zp56v
" is forbidden: exceeded quota: dev-quota, requested: pods=1, used: pods=5, limited: pods=5
  Normal   SuccessfulCreate  26s                replicaset-controller  Created pod: nginx-deployment-6bcdfcf6cc-r4rtc
  Warning  FailedCreate      26s                replicaset-controller  Error creating: pods "nginx-deployment-6bcdfcf6cc-vsfl2
" is forbidden: exceeded quota: dev-quota, requested: pods=1, used: pods=5, limited: pods=5
  Warning  FailedCreate      26s                replicaset-controller  Error creating: pods "nginx-deployment-6bcdfcf6cc-99ssp
" is forbidden: exceeded quota: dev-quota, requested: pods=1, used: pods=5, limited: pods=5
  Warning  FailedCreate      26s                replicaset-controller  Error creating: pods "nginx-deployment-6bcdfcf6cc-j4s7h
" is forbidden: exceeded quota: dev-quota, requested: pods=1, used: pods=5, limited: pods=5
  Warning  FailedCreate      26s                replicaset-controller  Error creating: pods "nginx-deployment-6bcdfcf6cc-dfj97
" is forbidden: exceeded quota: dev-quota, requested: pods=1, used: pods=5, limited: pods=5
  Warning  FailedCreate      26s                replicaset-controller  Error creating: pods "nginx-deployment-6bcdfcf6cc-c7fd8
" is forbidden: exceeded quota: dev-quota, requested: pods=1, used: pods=5, limited: pods=5
  Warning  FailedCreate      26s                replicaset-controller  Error creating: pods "nginx-deployment-6bcdfcf6cc-r87lh
" is forbidden: exceeded quota: dev-quota, requested: pods=1, used: pods=5, limited: pods=5
  Warning  FailedCreate      26s                replicaset-controller  Error creating: pods "nginx-deployment-6bcdfcf6cc-5vd5b
" is forbidden: exceeded quota: dev-quota, requested: pods=1, used: pods=5, limited: pods=5
  Warning  FailedCreate      25s                replicaset-controller  Error creating: pods "nginx-deployment-6bcdfcf6cc-lvcn7
" is forbidden: exceeded quota: dev-quota, requested: pods=1, used: pods=5, limited: pods=5
  Warning  FailedCreate      2s (x12 over 25s)  replicaset-controller  (combined from similar events): Error creating: pods "n
ginx-deployment-6bcdfcf6cc-v74kv" is forbidden: exceeded quota: dev-quota, requested: pods=1, used: pods=5, limited: pods=5
```

```
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# kubectl describe ns dev
Name:         dev
Labels:       area=dev
              kubernetes.io/metadata.name=dev
Annotations:  <none>
Status:       Active

Resource Quotas
  Name:      dev-quota
  Resource  Used  Hard
  --------  ---   ---
  cpu       0     2
  memory    0     3G
  pods      0     5
  services  0     3

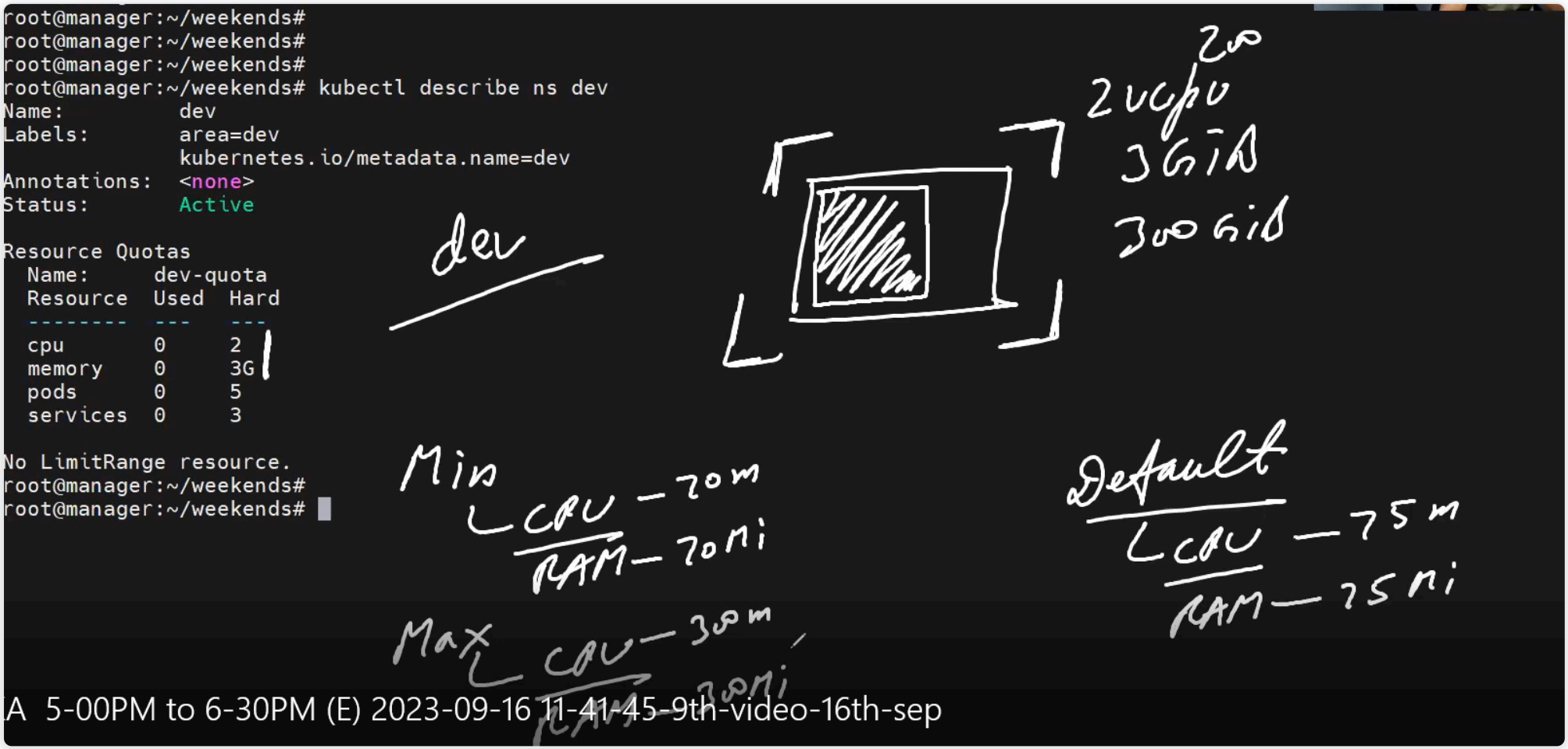
No LimitRange resource.
root@manager:~/weekends#
root@manager:~/weekends#
```



```
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# kubectl describe ns dev
Name:         dev
Labels:       area=dev
              kubernetes.io/metadata.name=dev
Annotations:  <none>
Status:       Active

Resource Quotas
  Name:     dev-quota
  Resource  Used  Hard
  --------  ---   ---
  cpu       0     2
  memory    0     3G
  pods      0     5
  services  0     3

No LimitRange resource.
root@manager:~/weekends#
root@manager:~/weekends#
```



A 5-00PM to 6-30PM (E) 2023-09-16 11-41-45-9th-video-16th-sep

```yaml
apiVersion: v1
kind: LimitRange
metadata:
  name: def-cpu-mem-limit
  namespace: dev
spec:
  limits:
  - default:
      cpu: 111m
      memory: 99Mi
    defaultRequest:
      cpu: 101m
      memory: 91Mi
    max:
      cpu: 200m
      memory: 100Mi
    min:
      cpu: 100m
      memory: 90Mi
    type: Container
```

```
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# kubectl describe ns dev
Name:         dev
Labels:       area=dev
              kubernetes.io/metadata.name=dev
Annotations:  <none>
Status:       Active

Resource Quotas
  Name:     dev-quota
  Resource  Used  Hard
  --------  ---   ---
  cpu       0     2
  memory    0     3G
  pods      0     5
  services  0     3

Resource Limits
 Type       Resource  Min   Max   Default Request  Default Limit  Max Limit/Request Ratio
 ----       --------  ---   ---   ---------------  -------------  -----------------------
 Container  cpu       100m  200m  101m             111m           -
 Container  memory    90Mi  100Mi  91Mi            99Mi           -
root@manager:~/weekends#
```

```
Resource Limits
 Type       Resource  Min   Max    Default Request  Default Limit  Max Limit/Request Ratio
 ----       --------  ---   ---    ---------------  -------------  -----------------------
 Container  cpu       100m  200m   101m             111m           -
 Container  memory    90Mi  100Mi  91Mi             99Mi           -
root@manager:~/weekends# vim mypod.yaml
root@manager:~/weekends# vim mypod.yaml
root@manager:~/weekends#
root@manager:~/weekends# kubectl create -f mypod.yaml
Error from server (Forbidden): error when creating "mypod.yaml": pods "webapp" is forbidden: [minimum memory usage per Contain
er is 90Mi, but request is 80Mi, minimum cpu usage per Container is 100m, but request is 50m]
root@manager:~/weekends#
```